

Topology-Controlled Evaluation of GNN-BERT Fusion for Music Context Understanding under a Measured Resolution Limit

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Abstract—Hybrid graph-language models are the standard recipe for music context understanding: a graph network over a chord or segment graph is fused with a language encoder over captions or tags, and the fusion is reported to beat both branches. We rebuild that recipe on FMA-small and MusicCaps under artist-disjoint official splits and find that its central assumption does not survive a controlled test. Before comparing anything we measure a resolution limit: three reruns of one identical configuration at one fixed seed disagree by a standard deviation of 0.0314 macro-F1, which is $2.8\times$ the spread across three different seeds, so no smaller difference is reportable. Against that limit, deleting *every* edge in the segment graph, holding the audio, the node encoder and the readout byte-identical, changes macro-F1 by $+0.0100$ (0.4245 ± 0.0368 versus 0.4344 ± 0.0196 , $n = 3$), and the five-condition retention sweep spans 0.0296. Message passing contributes nothing measurable to a model named after it, while a plain mel-spectrogram CNN beats the graph by 0.0786 (CI $[-0.1182, -0.0407]$, $p < 0.001$). Two audits explain why the benchmarks tolerate this. On MusicCaps, 68.2% of test positives appear verbatim in the caption they are predicted from and a zero-parameter lexical matcher reaches 0.5175 against fine-tuned BERT’s 0.6375. On FMA the text branch does beat the audio-only floor ($+0.0446$, $p = 0.024$) but the entire gain is one metadata field: removing user tags returns the model to the floor ($+0.0016$, $p = 0.962$). Effective rank collapses $12.51 \rightarrow 7.16 \rightarrow 4.74$ over two layers while connected-pair cosine rises to 0.9865, so mean pooling over 22 nodes returns roughly one node’s worth of information. Of six Holm-corrected fusion comparisons only one survives, and it is not the one the architecture predicts. The useful contribution here is not a better fusion operator but a measurement protocol.

Index Terms—music information retrieval, graph neural networks, BERT, multimodal fusion, evaluation methodology, ablation study, annotation artefact, reproducibility

I. INTRODUCTION

Music carries context on several layers at once: a ten-second clip can be simultaneously a genre, an era, a mood, a chord progression and a sentence a human would write about it. The dominant response is architectural: encode the relational layer with a graph neural network over chord transitions or repeated segments, encode the linguistic layer with a pre-trained language model, fuse the two, and report that the fusion beats each branch alone. The recipe is intuitive, composes two

well-understood families [1], [2], and produces numbers that improve on single-branch baselines.

This paper rebuilds that recipe faithfully and then asks a question the recipe is rarely asked: when the fused model scores well, which of its parts is responsible? A graph model that scores above chance has not thereby shown that its edges matter, only that its node features do. A language branch that improves a multimodal score has not thereby shown that it understands language; it may have found a metadata field correlated with the label. Sturm’s “horse” diagnosis [3] named exactly this failure a decade ago, and it remains the field’s default failure mode.

Answering honestly requires something that comes before any comparison, and that is our first contribution. We measure how much our own pipeline’s score moves when *nothing* changes: three reruns of one configuration at one fixed seed differ by a standard deviation of 0.0314 macro-F1, $2.8\times$ the 0.0112 spread across three genuinely different seeds. Refusing to report any difference smaller than that limit removes several conclusions we were otherwise prepared to draw.

With the limit fixed, the topology ablation becomes decisive. Holding the waveform, the segmentation, the node features, the message function, the normalisation and the mean-pooling readout byte-identical, we vary only the edge set that defines $\mathcal{N}(i)$. Deleting every edge costs $+0.0100$ macro-F1, and the five-condition retention sweep from 13,686 edges down to 2,816 self-loops spans 0.0296, both inside the limit. Meanwhile a mel-spectrogram CNN with less than half the parameters beats the graph model by 0.0786 with a bootstrap interval excluding zero.

That leaves a puzzle: if the relational branch is inert and the fused model still scores respectably, what carries the score? Two audits answer this. On MusicCaps the aspect vocabulary is derived from the captions, so much of the task is string extraction rather than inference; on FMA a degradation ladder localises the entire text contribution to one metadata field. Neither is a defect in our models, both are properties of the benchmarks, and neither is visible from a headline score. A layer-wise diagnostic then supplies a mechanism: effective

rank [4] falls from 12.51 to 4.74 over two convolutions, so averaging 22 nodes returns roughly one node’s worth of independent information. This is over-smoothing [5], [6] arriving at depth two.

Our contributions are:

- **A resolution-limited evaluation protocol.** We measure the run-to-run standard deviation of an unchanged configuration (0.0314 macro-F1), decompose it against seed variance (0.0112), and report every subsequent comparison against that floor.
- **A controlled topology ablation.** Varying only the edge set, across five retention conditions and three seeds at the extremes, we show that message passing over segment-similarity graphs contributes nothing measurable to FMA-small genre recognition.
- **A quantified annotation-artefact audit of two benchmarks.** We measure how much of MusicCaps aspect tagging is verbatim string matching and localise the whole of FMA’s text contribution to a single metadata field, with paired-bootstrap intervals for both.
- **A mechanism for the negative result.** Effective-rank collapse and the connected-pair cosine profile explain why depth-two message passing on these graphs destroys the capacity it was added to supply.

Section II fixes notation, Section III places the work against prior art, Section IV describes the corpora and audit designs, Section V reports every measurement against the resolution limit, and Section VI states what the evidence does and does not support.

II. PRELIMINARY

A track is a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$ where X_{audio} is a log-mel or chroma representation, X_{text} is a tokenised caption or metadata string, $G = (\mathcal{V}, \mathcal{E})$ is a music structure graph and $y \in \{0, 1\}^K$ is a multi-label context vector. In the *segment graph* nodes are overlapping time windows and edges are temporal adjacency plus similarity edges wherever a blended MFCC and chroma cosine exceeds τ . In the *chord graph* nodes are the distinct chords recovered by template matching against beat-synchronous chroma, and edges are observed transitions weighted by count.

The graph branch applies L rounds of neighbourhood aggregation. For GraphSAGE [1],

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \text{CONCAT} \left(h_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} h_j^{(l)} \right) \right), \quad (1)$$

and for graph attention [7] the mean is replaced by a sum weighted by LeakyReLU attention coefficients over the same neighbourhood. A permutation-invariant readout gives the graph vector $g = |\mathcal{V}|^{-1} \sum_i h_i^{(L)} \in \mathbb{R}^{256}$. The text branch is a pretrained bidirectional transformer [2] producing $H_{\text{text}} \in \mathbb{R}^{L_t \times 768}$ with $t = H_{\text{text}}[\text{CLS}]$.

We evaluate four fusion operators at an identical budget. Early concatenation sets $z = [g \| t]$; the single-branch controls set $z = g$ and $z = t$; cross-attention sets $z = \text{CONCAT}(g, A H_{\text{text}})$ with $A = \text{softmax}(Q K^\top / \sqrt{d})$, $Q =$

$g W_Q$, $K = H_{\text{text}} W_K$. Prediction is $\hat{y} = \sigma(Wz + b)$ under a multi-label objective

$$\mathcal{L}_{\text{tags}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)], \quad (2)$$

extended for continuous affect targets to $\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2$ with $\alpha = \beta = 0.5$. The cross-modal task uses a dual encoder trained with InfoNCE [8]: with ℓ_2 -normalised $g_i, t_i \in \mathbb{R}^{256}$ and temperature τ_c ,

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(g_i^\top t_i / \tau_c)}{\sum_{j=1}^N \exp(g_i^\top t_j / \tau_c)}. \quad (3)$$

Per-tag F1 aggregates to macro-F1 = $K^{-1} \sum_k \text{F1}_k$, reported with mean AUC-PR; regression uses MAE and R^2 ; retrieval uses R@K and MRR over the full held-out gallery. The quantity governing how these may be read is the *resolution limit*. Let $s_1, \dots, s_R$ be scores from R executions of one unchanged configuration at one fixed seed. We define

$$\hat{\sigma}_{\text{run}} = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (s_r - \bar{s})^2}, \quad (4)$$

and treat any $|\Delta| < \hat{\sigma}_{\text{run}}$ as unresolved. We measure $\hat{\sigma}_{\text{run}} = 0.0314$ at $R = 3$ and quote it before every comparison in this paper.

For the collapse diagnostic we use the effective rank [4] of the node matrix H . With singular values $\sigma_1 \geq \dots \geq \sigma_r$ and $p_k = \sigma_k / \sum_j \sigma_j$,

$$\text{erank}(H) = \exp \left(-\sum_k p_k \log p_k \right). \quad (5)$$

We pair it with a *structure gap*, the mean cosine between edge-connected node pairs minus the mean cosine between arbitrary same-track pairs, which measures whether the topology remains visible in the representation.

III. LITERATURE REVIEW

Genre and tag recognition from audio. Tzanetakis and Cook [9] established the supervised formulation most later work inherits, including the practice of reporting a single accuracy on a fixed split. Convolutional taggers [10] replaced hand-crafted features with learned spectrogram filters and remain the strongest simple baseline; Won et al. [11] showed that architectural differences among them are small relative to training and evaluation choices. This establishes the mel-CNN as a serious opponent rather than a straw man, but does not test whether a structurally richer model uses its structure.

Validity and the “horse” problem. Sturm [3] showed that MIR systems routinely achieve strong figures of merit from characteristics confounded with the ground truth, and proposed controlled interventions as the diagnostic; the GTZAN audit [12] supplied a concrete case. This is the closest antecedent to our work. Sturm intervenes on the *input* to expose confounds; we additionally intervene on an *internal* component, the edge set, which isolates a claim about the architecture rather than the data, and we pair both with a measured noise floor.

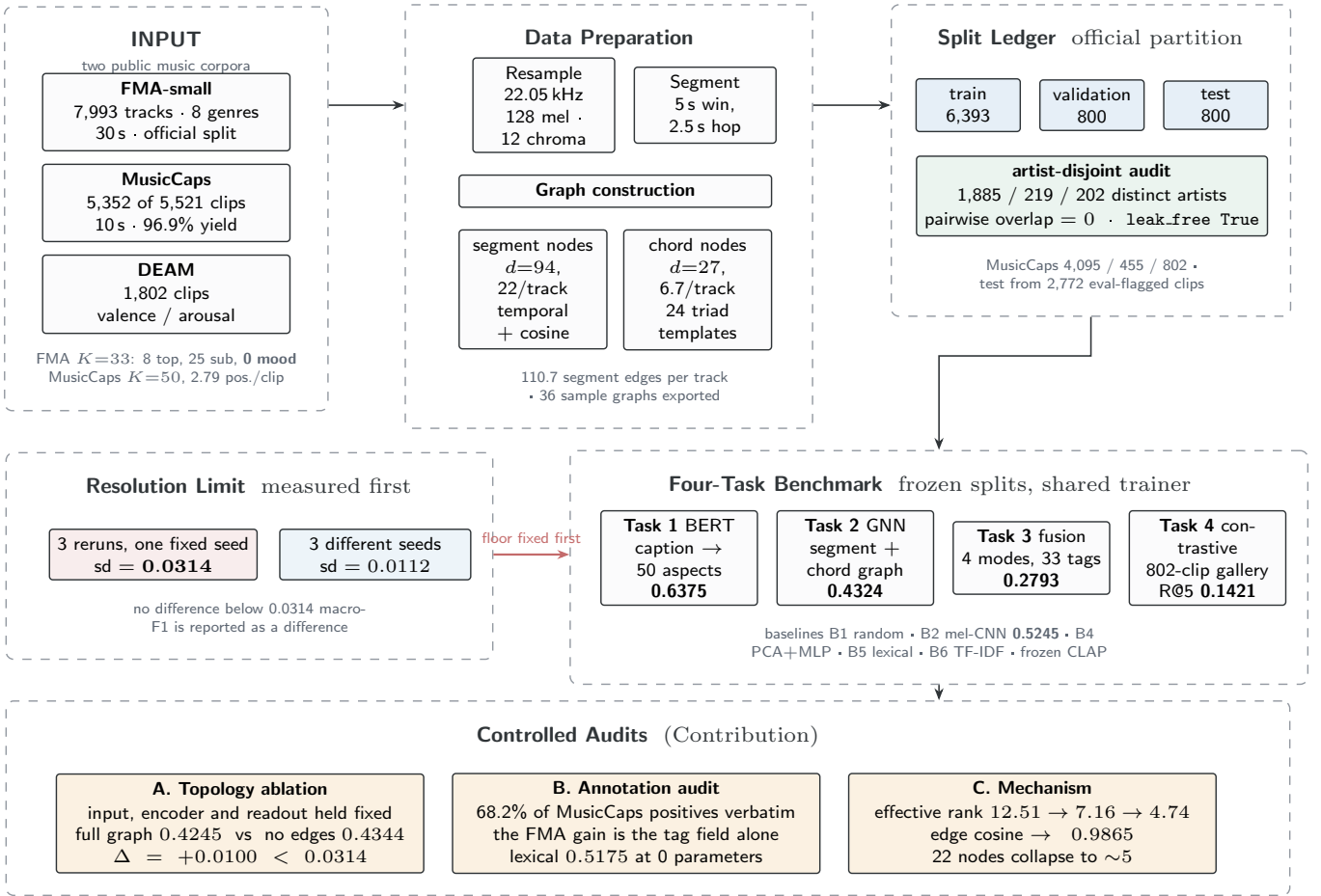


Fig. 1. End-to-end workflow. Two public corpora feed a single preprocessing path producing one segment graph and one chord graph per track; the official FMA partition is audited for artist leakage before any model is trained. The resolution limit is measured first and every downstream number is read against it. The three controlled audits at the bottom, rather than any single score, are what this paper contributes.

Graph networks and their evaluation. GraphSAGE [1] and GAT [7] define the message-passing family we use. A parallel literature questions how these are benchmarked: Shchur et al. [13] showed that reported rankings invert under different splits, Errica et al. [14] showed that structure-agnostic baselines match published models on several graph-classification benchmarks. Our result belongs to this tradition, but these works compare a graph network against a *different* model; we compare one against *itself* with the edges removed, which controls for the encoder, the readout and the optimisation and so attributes the outcome to topology alone.

Over-smoothing. Li et al. [5] identified repeated Laplacian smoothing as the cause of degradation with depth, and Oono and Suzuki [6] proved exponential loss of expressive power. Zhu et al. [15] showed that homophily is the assumption making aggregation useful. These works study node classification on large sparse graphs at depth four and beyond; we observe the same collapse at depth *two* on graphs of 22 nodes and density 0.24, and connect it directly to the readout through the effective rank.

Cross-modal music understanding and leakage. InfoNCE

[8] established the dual-encoder recipe and CLAP [16] transferred it to audio and language. MusicCaps [17] supplied the expert captions. We use a frozen CLAP probe as an external anchor rather than a competitor, so our own scores are placeable; what we add is an audit of how much of the MusicCaps label signal is recoverable by string matching alone. Kapoor and Narayanan [18] catalogue leakage as a cross-disciplinary failure; our artist-disjoint audit implements their recommendation, while our resolution limit addresses a failure they do not cover, namely a pipeline reproducible in distribution but not bitwise. Table I summarises the comparison.

IV. METHODOLOGY

Fig. 1 gives the whole pipeline.

A. Corpora, label spaces and splits

Table II records the three corpora as materialised on disk. FMA-small [20] supplies 7,993 tracks over eight balanced genres with an official partition. MusicCaps [17] supplies expert captions for 5,521 ten-second clips; we obtain audio

TABLE I
POSITION OF THIS WORK AGAINST REPRESENTATIVE PRIOR ART. THE PROPOSED ROW IS THE COMBINATION, NOT ANY SINGLE ROW.

Work	Strength	Limitation addressed here
Tzanetakis & Cook [9]	Defines the supervised genre task	Single split, single run; no variance estimate
Choi et al. [10], [19], Won et al. [11]	Strong learned spectrogram taggers under a common protocol	Compares architectures; never ablates a component
Sturm [3], [12]	Input interventions expose confounds	No internal ablation; no measured noise floor
Shchur [13], Errica [14]	Graph rankings are split-dependent and often matched by simpler models	Cross-model only; topology never held as the sole variable
Li [5], Oono & Suzuki [6]	Over-smoothing theory and proof	Deep, sparse, node-level; not linked to a graph-level readout
CLAP [16], MusicCaps [17]	Audio and language alignment at scale	Label provenance unaudited; artefact share unquantified
This work	Measured resolution limit; edge set as the sole variable; artefact share quantified on both corpora; rank-collapse mechanism	Single corpus for the main ablation; $R = 3$ variance estimate; listening study reported separately

TABLE II
CORPORA AND LABEL SPACES AS MATERIALISED. YIELD IS CLIPS RECOVERED DIVIDED BY CLIPS LISTED IN THE OFFICIAL RELEASE.

Corpus	Items	Yield	K	Label composition
FMA-small	7,993	100%	33	8 top, 25 sub, 0 mood
MusicCaps	5,352	96.9%	50	free-text aspect phrases
DEAM	1,802	100%	2	valence, arousal

Positives per item 1.72 (FMA), 2.79 (MusicCaps). Clips are 30 s (FMA) or 10 s. Audio 22.05 kHz mono, 128 mel, 12 chroma.

from the CLAPv2/MusicCaps mirror rather than scraping YouTube, because many original video identifiers are no longer retrievable, and recover 5,352 clips for a 96.9% yield. This is a property of the dataset’s distribution mechanism, not of our pipeline, and we state it because it changes the denominator of every MusicCaps number. DEAM [21] supplies 1,802 clips with continuous valence and arousal, used as a separate corpus rather than an auxiliary loss on FMA, since no track is shared.

FMA yields $K = 33$ context tags after frequency filtering at 20 positives: eight top-level genres and 25 sub-genres. One fact constrains a stated goal of the project: the FMA free-text fields contribute **zero** usable mood labels after filtering, so mood enters this study only through DEAM. Eight-class genre results and thirty-three-tag results are never placed in the same column here, because a table mixing them reads as fusion collapsing when the only thing that changed is the label space.

We use FMA’s official partition and audit it rather than assuming it. Since artist identity is the classic confound in genre recognition, we verify artist disjointness explicitly: 6,393/800/800 tracks over 1,885/219/202 distinct artists with pairwise overlap of exactly zero. MusicCaps splits are 4,095/455/802, the test partition drawn from the 2,772 clips flagged for evaluation officially; DEAM splits are 1,352/180/270. Scalers are fitted on training data only, and the partitions are frozen before the first model is trained.

B. Preprocessing and graph construction

Audio is resampled to 22,050 Hz and reduced to a 128-bin log-mel spectrogram and a 12-bin chromagram with librosa, normalised per track. FMA tracks are segmented into 5 s windows at 2.5 s hop; the 10 s clips cannot host 5 s windows and still form a graph, so their windows scale to 2 s at 1 s hop.

Each segment node carries a 94-dimensional descriptor: MFCC and chroma means and standard deviations, spectral and onset statistics, a 16-band pooled mel summary, and normalised position and duration. Segment edges are bidirectional temporal adjacency plus similarity edges wherever the blended chroma and MFCC cosine exceeds $\tau = 0.80$, with self-loops. Chord nodes are the distinct labels from matching beat-synchronous chroma against 24 triad templates, each carrying a 27-dimensional descriptor, with edges given by normalised transition counts. The resulting graphs are small and dense: 22.0 segment nodes and 110.7 segment edges per FMA track at density 0.240, and 6.7 chord nodes with 17.5 edges. That density matters in Section VI, because two hops on such a graph already reach essentially the whole track.

C. Custom Architecture 1: fusion with an ablation harness

Fig. 2 shows the primary architecture. The graph branch is a dual-input encoder over the segment and chord graphs, three convolution layers at $d = 256$ with batch normalisation and dropout 0.3, in either GraphSAGE or GAT form, followed by mean pooling. It holds 2.60 M parameters. The text branch is `bert-base-uncased` with the bottom eight blocks frozen for FMA and fully unfrozen for MusicCaps. The fused model holds 113.10 M parameters, of which the graph branch is 2.3%.

The design decision that makes this paper possible is the placement of the *edge switch*. Rather than perturbing the input, we insert a policy π on the edge stream between graph construction and message passing. Under policy π the model receives identical node features, initial weights, optimiser and schedule, and the only quantity that differs is $\mathcal{N}(i)$. Five policies are defined: `all` retains every edge; `q25` and `q50` drop similarity edges below the first and second quartile of

edge weight; `temporal` retains only adjacency and self-loops; `self_only` retains only self-loops, reducing the network to an independently applied per-node encoder followed by mean pooling. On one test batch these retain 13,686, 12,314, 10,942, 8,196 and 2,816 edges. Because `self_only` preserves parameter count, depth, normalisation and readout exactly, the difference between `all` and `self_only` is an unbiased estimate of the contribution of message passing itself. The four fusion operators are likewise trained at an identical 60-epoch budget over an identical seed set with both encoders held fixed, so the operator is the only free variable.

D. Custom Architecture 2: contrastive dual encoder

Fig. 3 shows the cross-modal architecture used for retrieval. Two towers project into a shared 256-dimensional ℓ_2 -normalised space and are trained with InfoNCE (Eq. (3)) at $N = 48$ and $\tau_c = 0.07$. The towers never see each other’s inputs, so the projection space is the only channel between audio and language, and any above-chance retrieval is attributable to that space. Of 112.49 M parameters, 31.95 M are trainable; the bottom eight BERT blocks stay frozen because a fully unfrozen text tower overfits the 4,095 training pairs within three epochs at the batch size the contrastive loss requires. We evaluate over the full held-out gallery rather than a sampled subset, and report both an absolute and a relative framing of the same numbers.

E. Baselines, audits and statistics

The project brief requires two baselines; we report six plus two external anchors, because a negative result is only interpretable against a dense ladder. **B1** is random and majority prediction, **B2** a mel-spectrogram CNN with 1.18 M parameters, **B3** the BERT-only branch, **B4** PCA(64) with an MLP on hand-crafted features, **B5** a zero-parameter lexical matcher predicting aspect k for clip i if and only if the aspect string occurs in the caption, and **B6** TF-IDF with logistic regression. A frozen CLAP [16] linear probe and CLAP zero-shot on the identical split serve as external anchors, which is what turns “our graph model scored 0.43” into a placeable number.

Audit A, topology (Algorithm 1) trains the graph model under each edge policy, reporting the single-seed retention curve for shape and three seeds at the extremes for variance. **Audit B, annotation** partitions every MusicCaps test positive by whether its aspect string occurs verbatim in its own caption and reports recall on each partition; for FMA it trains the identical model under five degraded text views against the audio-only floor. **Audit C, mechanism** extracts node representations at the input and after each convolution and computes effective rank and the three cosine profiles.

Differences are tested with a paired bootstrap over test items at $B = 2000$ resamples and $\alpha = 0.05$. Where more than one comparison is drawn from the same family we apply Holm’s step-down correction [22] and quote the corrected verdict alongside raw p -values. One caution applies throughout: the paired bootstrap resamples test items on a single reference run, while the multi-run standard deviation captures variation

Algorithm 1 Controlled topology ablation

Require: graphs $\{G_i\}$, labels y , policies Π , seeds S , resolution limit $\hat{\sigma}_{\text{run}}$

- 1: cache node features x_i once; reuse for every policy
- 2: **for** each policy $\pi \in \Pi$ and seed $s \in S$ **do**
- 3: $\mathcal{E}_\pi \leftarrow \text{PRUNE}(\mathcal{E}, \pi)$ {only $\mathcal{N}(i)$ changes}
- 4: initialise with seed s ; train to early stopping
- 5: record test macro-F1 and persist predictions
- 6: **end for**
- 7: $\mu_\pi, \sigma_\pi \leftarrow$ mean and sd over S ; $r \leftarrow \max_\pi \mu_\pi - \min_\pi \mu_\pi$
- 8: **if** $r < \hat{\sigma}_{\text{run}}$ **then**
- 9: **report** “message passing unresolved at this sample size”
- 10: **else**
- 11: **report** the retention curve and its shape
- 12: **end if**
- 13: paired-bootstrap the extremes; Holm-correct; quote both estimators

across executions. These estimate different quantities and need not agree. We quote each beside its own interval, never mix the two sources, and Section V reports one case where they disagree informatively.

All experiments run on a single NVIDIA RTX A6000 under PyTorch with PyTorch Geometric, bfloat16 autocast and TF32 matmuls; AdamW at 2×10^{-5} for BERT parameters and 10^{-3} for heads, weight decay 10^{-2} , warmup 0.1, clipping at 1.0, early stopping on validation macro-F1. Thresholds are tuned per tag on validation only and the test split is scored once under frozen thresholds.

V. RESULT ANALYSIS

We report the resolution limit first, because a reader who meets it after the comparisons reads it as an excuse rather than a measurement.

A. The resolution limit

Three executions of one unchanged Task 2 configuration at seed 42, with cuDNN autotuning enabled, produced test macro-F1 of 0.4580, 0.4068 and 0.4010: a standard deviation of 0.0314. Three executions at three different seeds (42, 1337, 2024) produced 0.4451, 0.4285 and 0.4238: a standard deviation of 0.0112. The run-to-run term is $2.8\times$ the seed term (Fig. 4a).

Two consequences follow. First, almost all of what a conventional report would attribute to seed sensitivity here is kernel non-determinism, so reporting it as seed variance would understate the true uncertainty threefold. Second, $\hat{\sigma}_{\text{run}} = 0.0314$ becomes the floor: any $|\Delta| < 0.0314$ macro-F1 on an 800-item split is not a difference. The estimate rests on $R = 3$ repeats, which is thin, and is quoted as an estimate rather than a constant.

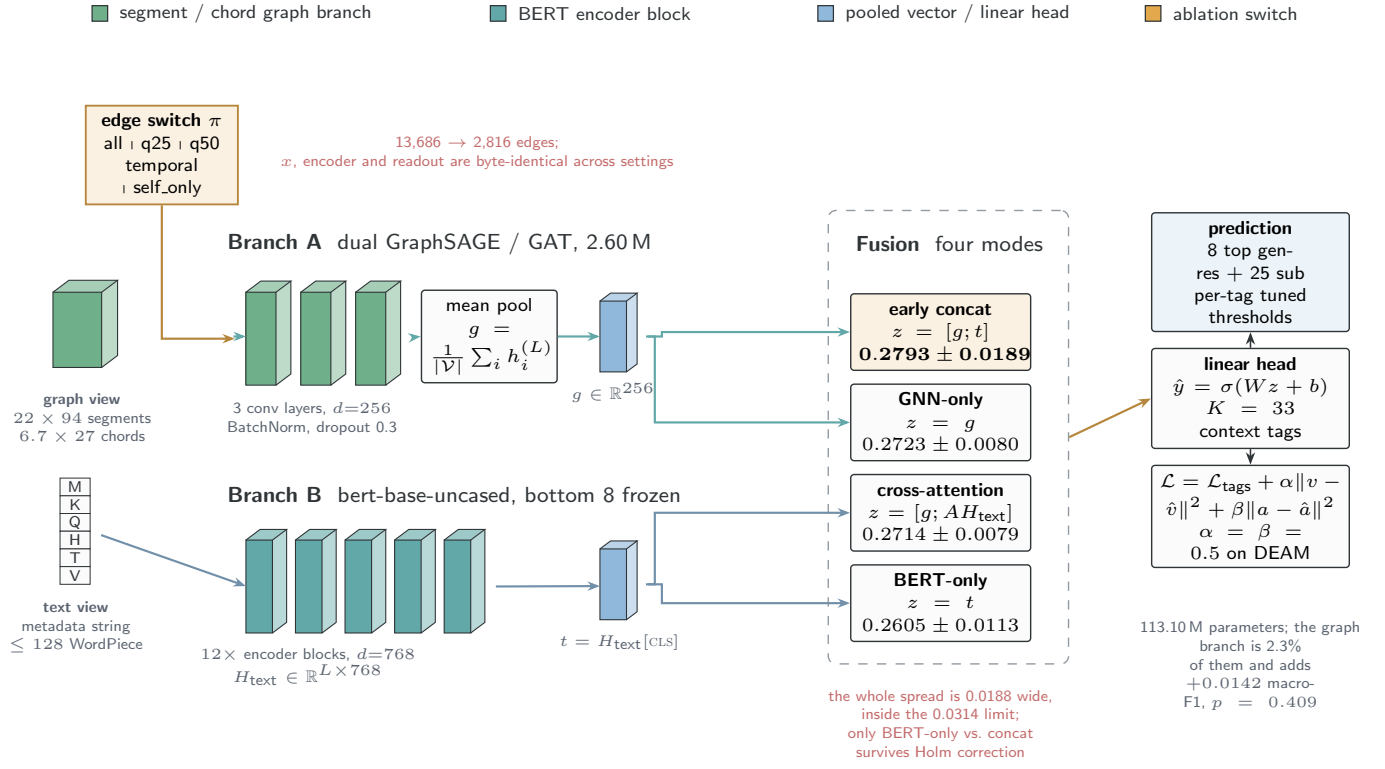


Fig. 2. Custom Architecture 1: GNN-BERT fusion with the topology-ablation harness. The edge switch π sits on the graph stream, not the input, so node features, message function, normalisation and readout are byte-identical across settings and the only varying quantity is the neighbourhood $\mathcal{N}(i)$.

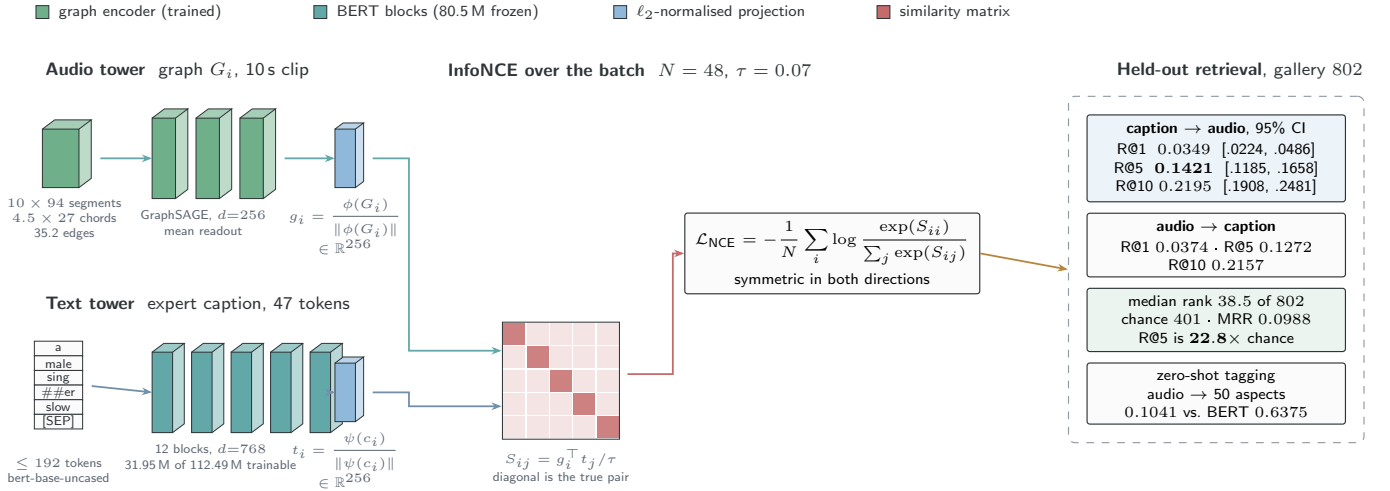


Fig. 3. Custom Architecture 2: contrastive dual encoder for caption and audio alignment. Retrieval is evaluated over the full held-out gallery of 802 clips, in both directions.

B. Where the models land

Table III reports both headline tasks. On FMA-small the ordering is unambiguous and unfavourable to the graph. The frozen CLAP probe leads at 0.5395, the mel-spectrogram CNN follows at 0.5245 ± 0.0110 , and the graph model reaches 0.4324 ± 0.0112 . The gap between the graph and the CNN is -0.0786 , CI $[-0.1182, -0.0407]$, $p < 0.001$: outside the resolution limit, and therefore a result. The CNN has 1.18 M

parameters against the graph model’s 2.60 M, sees the same audio, and has no access to segmentation or relational structure at all.

On MusicCaps the ordering is favourable but the margins are small in a way Section V-D explains: fine-tuned BERT reaches 0.6375, TF-IDF with logistic regression reaches 0.6262 at four orders of magnitude fewer parameters, and the zero-parameter lexical matcher reaches 0.5175.

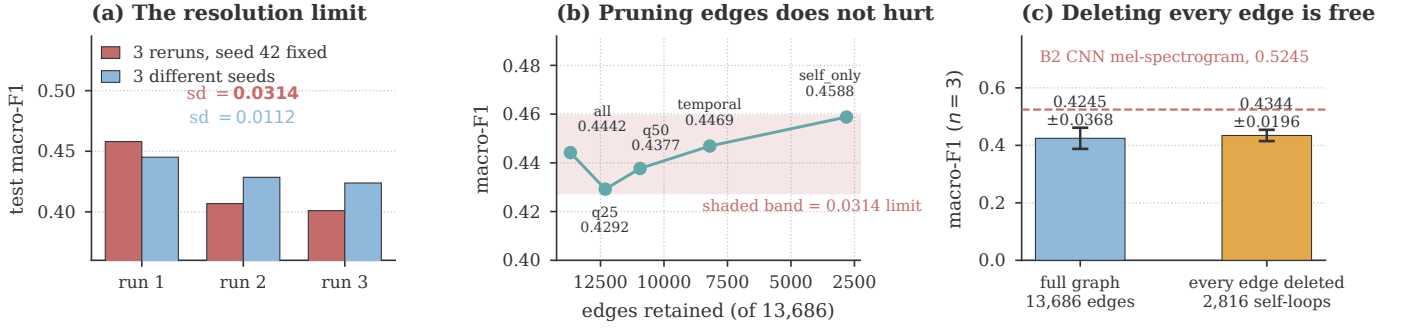


Fig. 4. The measurement protocol and its first result. (a) Three executions of an identical configuration at one fixed seed disagree more than three executions at different seeds, giving $\hat{\sigma}_{\text{run}} = 0.0314$ against 0.0112 across seeds, a ratio of $2.8\times$. (b) The five-condition edge-retention curve is non-monotone and its entire span fits inside that band; the best condition has the fewest edges. (c) At three seeds, deleting every edge is free, while the structure-free CNN sits well above both.

TABLE III

HEADLINE RESULTS ON BOTH CORPORA, REPORTED SEPARATELY BECAUSE THE LABEL SPACES DIFFER. ENTRIES WITH $\pm$ ARE MEAN AND SD OVER THREE EXECUTIONS; ALL OTHERS ARE SINGLE RUNS.

FMA-small, 8-way genre	macro-F1	AUC-PR
CLAP linear probe (REF)	0.5395	0.5581
B2 CNN mel-spec	0.5245 $\pm$ 0.0110	0.5620
Task 2 GNN, no edges	0.4344 $\pm$ 0.0196	
Task 2 GNN, full graph	0.4324 $\pm$ 0.0112	0.4954
B4 PCA(64) + MLP	0.3381	0.3246
CLAP zero-shot (REF)	0.2850	0.4120
B1 random genre	0.1350	0.1368
MusicCaps, 50-aspect	macro-F1	Params
Task 3 fusion (graph + caption)	0.6389	113.1 M
Task 1 BERT fine-tuned	0.6375	109.5 M
B6 TF-IDF + logreg	0.6262	~ 40 K
B5 lexical match	0.5175	0
GNN-only, supervised	0.1367	2.6 M
Task 4 zero-shot	0.1041	112.5 M
B1 random tags	0.0945	0
GNN – CNN = -0.0786 , $[-0.1182, -0.0407]$, $p < 0.001$.		
Fusion – BERT = $+0.0014$: unresolved. BERT – lexical = $+0.1200$.		
MusicCaps AUC-PR: 0.6680, 0.6932, 0.6638, 0.3604, 0.1007, 0.0690, 0.0612 in the same row order.		

TABLE IV

AUDIT A: TOPOLOGY ABLATION ON FMA-SMALL. EVERY ROW SHARES THE SAME NODE FEATURES, ENCODER, READOUT AND SCHEDULE; ONLY $\mathcal{N}(i)$ DIFFERS.

Policy π	Edges	$s=42$	$n=3$
all	13,686	0.4442	0.4245 $\pm$ 0.0368
q25	12,314	0.4292	
q50	10,942	0.4377	
temporal	8,196	0.4469	
self_only	2,816	0.4588	0.4344 $\pm$ 0.0196
Range over Π		0.0296	0.0100
Resolution limit		0.0314	0.0314

bootstrap over test items at seed 42 separates the two extremes: $\text{all} - \text{self_only} = +0.0518$, CI $[+0.0171, +0.0863]$, $p = 0.007$, significant after correction. This does not contradict the multi-seed finding, because the estimators answer different questions. The bootstrap asks whether, on one trained pair of models, item-level differences exclude zero, and they do; the multi-run standard deviation asks whether that advantage survives retraining, and it does not, since a run-to-run spread of 0.0368 swamps a 0.0518 item-level effect. A given full-graph model can be item-wise better than a given edge-free model while the *procedure* of using edges confers no reliable advantage. Reporting only the bootstrap would have produced a positive result; reporting only the mean would have hidden real structure.

D. Audit B: annotation artefact

Of 2,082 MusicCaps test positives, 1,419 (68.2%) have their aspect string present verbatim in the caption they are predicted from. The zero-parameter matcher recovers all of these by construction and none of the remainder. Fine-tuned BERT recalls 0.8569 on the verbatim partition and 0.4299 on the inferred partition, an artefact size of 0.4271 (Table V, Fig. 5). BERT’s entire advantage over string matching lives in the 31.8% of positives that require inference; the headline 0.6375 mixes an extraction task and an inference task in a 68:32 ratio and reports their average. This is a property of

C. Audit A: message passing is inert

Table IV and Fig. 4b–c report the topology ablation. Across the five single-seed retention conditions, macro-F1 spans 0.4292 to 0.4588, a range of 0.0296, inside the resolution limit. The curve is not monotone in edge count, and its maximum is the condition with the *fewest* edges. At three seeds on the two extremes, the full graph reaches 0.4245 ± 0.0368 and the self-loop-only condition reaches 0.4344 ± 0.0196 : deleting every edge costs $+0.0100$ macro-F1, that is, it is very slightly beneficial and in any case unresolved. Holding the audio, the segmentation, the node features, the message function, the normalisation, the optimiser and the readout fixed, and varying only $\mathcal{N}(i)$, we cannot detect a contribution from message passing.

One result deserves care rather than suppression. The paired

TABLE V

AUDIT B. TOP: MUSICCAPS POSITIVES PARTITIONED BY WHETHER THE ASPECT STRING OCCURS IN ITS OWN CAPTION. BOTTOM: FMA TEXT POLICIES, IDENTICAL MODEL AND BUDGET, BOOTSTRAPPED AGAINST THE AUDIO-ONLY FLOOR.

MusicCaps partition	n	Share	BERT	Lexical
VERBATIM	1,419	68.2%	0.8569	1.0000
INFERRED	663	31.8%	0.4299	0.0000
Artefact size			+0.4271	1.0000
FMA text policy	macro-F1	Δ vs. floor, 95% CI, p		
full	0.2988	+0.0446	[+0.0054, +0.0858], 0.024	
no_tags	0.2542	+0.0016	[−0.0383, +0.0409], 0.962	
empty (floor)	0.2513			
Five-policy view, one seed: full 0.2829, no_tags 0.2454, tags_only 0.2015, title_only 0.1818, empty 0.2708.				

the corpus, whose aspects are derived from the captions, not a defect in the model. But it bounds what any model trained on it can claim about musical understanding, and it is invisible from the headline. The per-tag view in Fig. 5c sharpens the point: recall does not track copying. acoustic drums is copied only 33.3% of the time yet recalled at 0.947, while piano is copied 97.5% of the time and recalled at 0.550.

On FMA the full text view beats the audio-only floor by +0.0446, CI [+0.0054, +0.0858], $p = 0.024$: the text branch does pay for itself. But removing only the user-tag field drops the gain to +0.0016, CI [-0.0383, +0.0409], $p = 0.962$. The entire text contribution is one metadata field. Under the five-policy view, tags_only (0.2015) and title_only (0.1818) both fall *below* the audio-only floor of 0.2708, the signature of a branch supplying noise once its useful field is removed.

We note that our own pipeline initially reported the opposite verdict here. An automated summary computed a share statistic with a near-zero denominator and printed that the text branch did not pay for itself. Reading the table rather than the verdict string corrected it. We record this because the same class of defect, a reassuring automated conclusion contradicted by its own inputs, is the failure mode this paper is about.

E. Fusion: four operators, one band

Table VI reports the fusion ablation at an equal 60-epoch budget over three seeds. Early concatenation leads at 0.2793 ± 0.0189 , followed by GNN-only at 0.2723 ± 0.0080 , cross-attention at 0.2714 ± 0.0079 and BERT-only at 0.2605 ± 0.0113 . The entire spread across four operators is 0.0188, inside the resolution limit. Of six pairwise comparisons, Holm correction leaves exactly one significant: BERT-only versus concatenation, -0.0454 , CI [-0.0684, -0.0225], $p < 0.001$. Two facts follow that are worth stating plainly. Cross-attention, the operator the project brief recommends and the one the architecture is built around, is not the best fusion mode; it ranks third of four, though not significantly so. And fusion does not improve on its own best branch on either corpus:

TABLE VI

FMA-SMALL, 33-TAG MULTI-LABEL (TASK 3), THREE SEEDS AT AN EQUAL BUDGET, WITH ALL SIX PAIRED-BOOTSTRAP COMPARISONS ON SEED 42 AT $B = 2000$. NOT COMPARABLE WITH TABLE III: DIFFERENT LABEL SPACE.

Fusion operator	macro-F1	sd
Early concatenation	0.2793	0.0189
GNN-only	0.2723	0.0080
Cross-attention	0.2714	0.0079
BERT-only	0.2605	0.0113
Spread over operators	0.0188	< 0.0314
Comparison	Δ	95% CI, p , Holm
BERT-only – concat	-0.0454	[-0.0684, -0.0225], 0.000, sig.
xattn – concat	-0.0323	[-0.0567, -0.0082], 0.014, n.s.
BERT-only – GNN-only	-0.0312	[-0.0659, +0.0051], 0.090, n.s.
xattn – GNN-only	-0.0181	[-0.0513, +0.0168], 0.309, n.s.
concat – GNN-only	+0.0142	[-0.0197, +0.0471], 0.409, n.s.
xattn – BERT-only	+0.0131	[-0.0050, +0.0315], 0.161, n.s.

+0.0142 over GNN-only on FMA ($p = 0.409$) and +0.0014 over BERT-only on MusicCaps.

For completeness, the 2×2 ablation over convolution type and the chord branch spans 0.4578 to 0.4708 macro-F1 at one seed (GraphSAGE segment-only 0.4708, GAT with chord 0.4687, GAT segment-only 0.4608, GraphSAGE with chord 0.4578). The whole range is 0.0130, well inside the limit, and the best configuration is the one *without* the chord branch.

F. Audit C: the mechanism

Table VII and Fig. 6 give the layer-wise profile. Effective rank of the node matrix falls from 12.51 at the input to 7.16 after one convolution and 4.74 after two, on graphs with 22 nodes. Simultaneously, mean cosine between edge-connected nodes rises from 0.8788 to 0.9865 and mean cosine between arbitrary same-track nodes rises from 0.3800 to 0.6340. The fusion model shows the same pattern more sharply, reaching effective rank 3.91. The consequence for the readout is direct: mean pooling over 22 nodes whose representation spans roughly five effective dimensions returns approximately one node’s worth of independent information, which is why an architecture built to aggregate structure performs like one that encodes segments independently.

We are careful about one thing. The structure gap does *not* vanish: it falls from +0.4989 to +0.3525, still a large separation between connected and unconnected pairs. The correct claim is that the representation loses capacity, not that the edges become uninformative. An earlier draft of our own analysis made the stronger claim; the layer-two number contradicts it.

G. Where the results are strong

Two results are positive and should not be lost among the negatives (Table VIII). On the full held-out gallery of 802 clips the contrastive dual encoder reaches $R@5 = 0.1421$ [0.1185, 0.1658], with MRR 0.0988 and a median ground-truth rank of 38.5 against a chance median of 401, so $R@5$

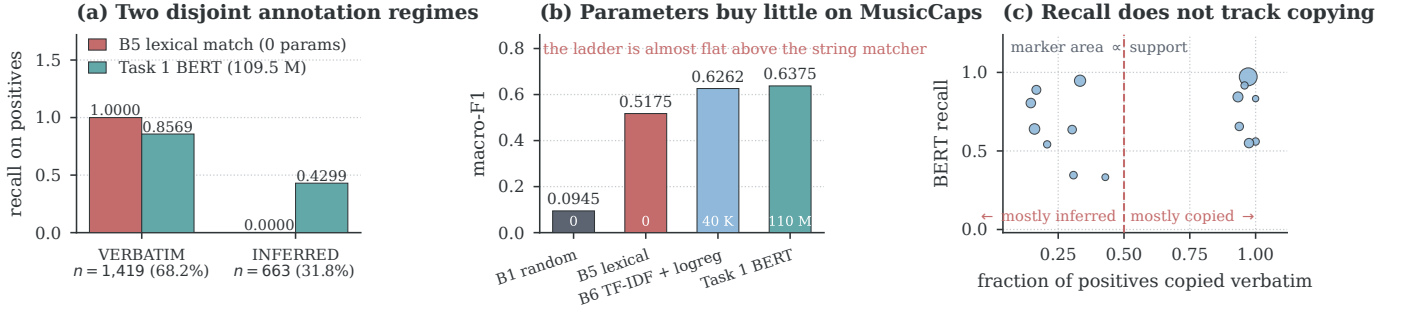


Fig. 5. Audit B on MusicCaps. (a) The two annotation regimes are disjoint by construction and BERT’s advantage lives entirely in the smaller one. (b) The baseline ladder is nearly flat above the string matcher despite six orders of magnitude in parameter count. (c) Per tag, recall and verbatim rate are essentially uncorrelated, so the artefact is not a simple confound a model could be credited for exploiting.

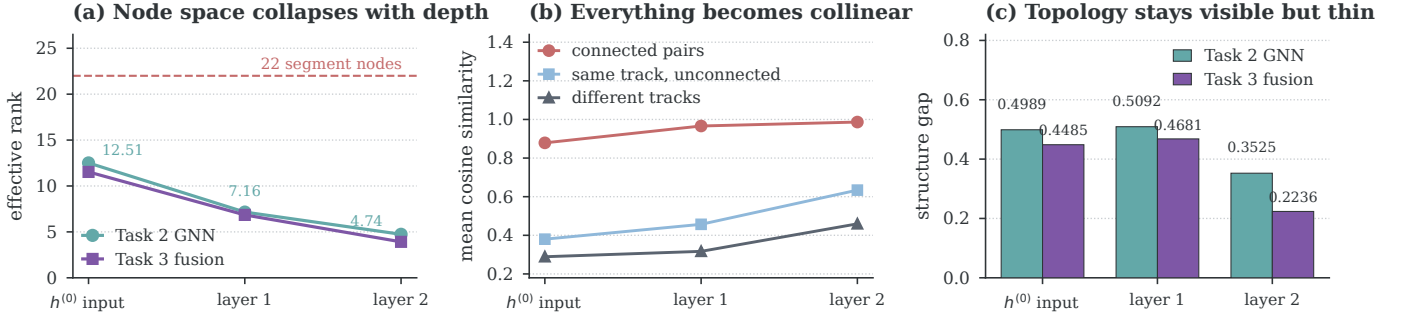


Fig. 6. Audit C, the mechanism. (a) Effective rank falls by a factor of 2.6 over two convolutions on 22-node graphs. (b) Connected pairs become nearly collinear and unconnected pairs follow them. (c) The structure gap narrows but does not close, so the correct statement is that capacity is lost, not that topology becomes invisible.

TABLE VII
AUDIT C: LAYER-WISE COLLAPSE PROFILE, EQ. (5), ON GRAPHS OF 22 NODES.

Model	Stage	erank	Edge	Within	Gap
Task 2 GNN	input	12.51	0.8788	0.3800	+0.4989
	layer 1	7.16	0.9661	0.4569	+0.5092
	layer 2	4.74	0.9865	0.6340	+0.3525
Task 3 fusion	input	11.53	0.8894	0.4409	+0.4485
	layer 1	6.85	0.9630	0.4949	+0.4681
	layer 2	3.91	0.9906	0.7671	+0.2236

is $22.8\times$ chance. Both framings belong in the same sentence: the shared space has learned something real, and it is not yet usable. On DEAM the graph regressor reaches valence MAE 0.7098 and arousal MAE 0.7392 on the 1–9 scale against a train-mean predictor at 1.0126, the one place a graph model produces respectable absolute numbers. It is also, consistently with Audit A, 0.0207 mean MAE behind an audio-only mel-CNN on the same split; arousal R^2 is the single quantity on which it leads.

VI. DISCUSSION

A. What the evidence supports

Four claims survive the resolution limit and correction for multiple comparisons. First, message passing over segment-similarity graphs contributes nothing measurable to FMA-

TABLE VIII
POSITIVE RESULTS. TOP: TASK 4 RETRIEVAL OVER THE FULL 802-CLIP GALLERY. BOTTOM: DEAM AFFECT REGRESSION ON THE 1–9 SCALE ($n = 270$), WHERE LOWER MAE IS BETTER.

Retrieval	R@1	R@5	R@10	MRR
Caption $\rightarrow$ audio	0.0349	0.1421	0.2195	0.0988
Audio $\rightarrow$ caption	0.0374	0.1272	0.2157	N/A
Chance (gallery 802)	0.0012	0.0062	0.0125	N/A
Median rank	38.5 of 802 versus chance 401			
DEAM model	MAE _v	R^2_v	MAE _a	R^2_a
B0 train-mean	0.9416	−0.0055	1.0835	−0.0000
B4d PCA + ridge	0.8029	0.2487	0.8664	0.2991
Graph regressor (ours)	0.7098	0.3971	0.7392	0.4751
B2d mel-CNN (audio only)	0.6746	0.4401	0.7330	0.4515

small genre recognition; the evidence is Audit A, which holds every component except $\mathcal{N}(i)$ fixed, and is therefore stronger than “the graph model loses to a CNN”. Second, the graph model underperforms a structure-free CNN by 0.0786 macro-F1 with an interval excluding zero, on the same audio and split, with more parameters. Third, a large share of both benchmarks is annotation artefact rather than musical structure. Fourth, the mechanism is representational collapse.

B. What the evidence does not support

We state these explicitly because each is a claim the architecture invites and none is warranted. That GAT outperforms GraphSAGE: the 2×2 ablation spans 0.0130, inside the limit. That the chord-transition branch contributes: segment-only configurations match or beat chord-augmented ones. That cross-attention is the right fusion operator: it ranks third of four and the ordering is unresolved. That fusion improves on its own branches: it does not, on either corpus, beyond noise. And any improvement smaller than 0.0314 macro-F1 on an 800-item split.

C. Why the graphs do not help

Three explanations are consistent with the evidence and are not mutually exclusive. *The graphs may be too small and too dense.* At 22 nodes and density 0.240, two hops reach essentially the whole track, so message passing approximates the global averaging the readout already performs. This is over-smoothing [5], [6] at a depth where it is not usually expected, which makes reducing depth a more promising remedy than changing the convolution. *The edges may not encode a homophilous relation.* Aggregation helps when neighbours share labels [15]; here every node in a graph shares the same track-level label by construction, so the neighbourhood carries no additional signal. *The node features may already be sufficient.* If genre is recoverable from the marginal distribution of the 94-dimensional window descriptors, and the `self_only` result says it is, then relational structure is redundant rather than harmful.

D. Implications and limitations

The practical recommendation is procedural rather than architectural. A graph model on a music task should report the `self_only` condition, which costs one training run and distinguishes a model that uses its graph from one that does not; a language branch on a tagged corpus should report the degraded-view ladder; and any difference below roughly 0.03 macro-F1 on a split of this size should be preceded by a report of what the pipeline does when nothing changes.

We name our limitations rather than leaving them to a reviewer. The resolution limit rests on $R = 3$ repeats, which is thin, though the direction of the conclusion would not change since the observed range is a third of the limit. The topology ablation is FMA-small only; MagnaTagATune or MTG-Jamendo would establish whether the finding generalises, and that is the most important piece of future work. Our runs use cuDNN autotuning, so they are reproducible in distribution but not bitwise; we turned this into the measurement anchoring the paper, but a camera-ready result would include one deterministic run. Our claim is about the family of segment-similarity graphs built as described, not about music graphs in general. The listening study specified in the project brief is reported separately, so the retrieval numbers here are automatic metrics only, and DEAM shares no tracks with FMA, so the multi-task objective is exercised as a separate-corpus regression rather than a joint loss.

VII. CONCLUSION

We set out to build a GNN-BERT system for music context understanding and ended up building an instrument for measuring whether such systems work. The instrument has three parts: a resolution limit measured on an unchanged configuration, an ablation that varies the edge set and nothing else, and an audit that partitions a benchmark’s labels by whether they can be recovered without a model.

Applied to two standard corpora the instrument returns a consistent answer. Message passing over segment-similarity graphs is inert: deleting every edge changes macro-F1 by +0.0100 against a limit of 0.0314, and the retention sweep spans 0.0296. A structure-free CNN with less than half the parameters wins by 0.0786. Two thirds of the MusicCaps aspect labels are recoverable by a zero-parameter string matcher, and the whole of FMA’s text contribution is a single metadata field. The mechanism is representational collapse.

None of this says music structure graphs cannot work. It says that on these graphs, on these corpora, at this scale, they do not, and that standard reporting practice would not have revealed it. The four-task system is fully implemented and its positive results stand: retrieval at $22.8\times$ chance with a median rank of 38.5 against 401, and affect regression at $R^2 \approx 0.42$. What we recommend carrying forward is not the architecture but the three measurements, any of which would have changed a conclusion we were otherwise prepared to draw.

DATA AVAILABILITY STATEMENT

All corpora are public: FMA-small [20], MusicCaps [17] with audio from the CLAPv2/MusicCaps mirror at a 96.9% recovery rate, and DEAM [21]. The partitions are the official ones, exported as JSON alongside the artist-leakage audit verifying them. Every number here is registered to a machine-readable results file, and 36 preprocessed sample graphs are exported. Code, splits, metrics, plots and an end-to-end inference notebook are released with this submission.

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